Name

Mathematics Extension 1 Year 12

Sum and Product of Zeroes

Book 1

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

ME1-11-02 applies the remainder and factor theorem and sums and products of zeroes to solve problems involving polynomials

Sums and products of zeroes of polynomials

- Prove that if a quadratic $P(x) = ax^2 + bx + c$ has zeroes α and β then $\alpha + \beta = -\frac{b}{a}$, the sum of the zeroes, and $\alpha\beta = \frac{c}{a}$, the product of the zeroes
- Prove that if a cubic $P(x) = ax^3 + bx^2 + cx + d$ has three zeroes α, β, γ, then $\alpha + \beta + \gamma = -\frac{b}{a}$, $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$ and $\alpha\beta\gamma = -\frac{d}{a}$
- Prove that if a quartic $P(x) = ax^4 + bx^3 + cx^2 + dx + e$ has four zeroes α, β, γ, δ then $\alpha + \beta + \gamma + \delta = -\frac{b}{a}$, $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\alpha = \frac{c}{a}$, $\alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta = -\frac{d}{a}$ and $\alpha\beta\gamma\delta = \frac{e}{a}$
- Use the formulas for the sums and products of zeroes to solve problems involving zeroes and coefficients of quadratic, cubic and quartic polynomials

Sum and Product of Roots (Quadratic)

Sum and Product of Zeroes for a Quadratic Equation

Consider this polynomial.

$$P(x) = (x-2)(x-3)(x-5)(x-7) = x^4 - 17x^3 + 101x^2 - 247x + 210$$

The coefficients of the expanded polynomial are: $1, -17, 101, -247, 210$
These must come from $2, 3, 5, 7$. But can we get them quickly *without* expanding?

Try adding all the numbers: $2 + 3 + 5 + 7 = \ldots\ldots\ldots$

Try multiplying all the numbers: $2(3)(5)(7) = \ldots\ldots\ldots$

If we can find a formula relating the coefficients and the zeroes, we can factorise and expand polynomials very quickly.

Let's start with a quadratic function $f(x) = ax^2 + bx + c$,

The **zeroes, or roots,** of the function are when $f(x) = \ldots\ldots\ldots$
This corresponds to the ……-intercepts of the function.

To find the zeroes, we can use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The **sum of the zeroes** will be:

The **product of the zeroes** will be:

Alternate proof

Let $ax^2 + bx + c = 0$, we can make this polynomial monic by $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$
suppose it can be factorised to $P(x) = (x-\alpha)(x-\beta)$.
By expanding $P(x) = (x-\alpha)(x-\beta)$, equate coefficients to determine $\alpha + \beta$ and $\alpha\beta$.

ⓘ Sum and Product of Zeroes (Quadratic)

For a **quadratic** $P(x) = ax^2 + bx + c$ with zeroes (roots) α and β:

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta = \frac{c}{a}$$

🔍 Sum and Product of Zeroes for Quadratics

By factorising, test that the sum and product of zeroes formula work for these quadratic functions.

a $P(x) = x^2 + 7x + 10$

$P(x) = (\quad)(\quad)$

$\alpha + \beta =$

$\alpha\beta =$

b $P(x) = x^2 + 3x - 10$

c $P(x) = 3x^2 + 21x + 30$

d $P(x) = 2x^2 + 5x - 12$

Example Apply sum and product of zeroes formulas for quadratics

Worked Example	Your Turn
If α and β are the roots of $2x^2 - 6x + 1 = 0$, find:	If α and β are the roots of $2x^2 + 3x + 5 = 0$, find:
$\alpha + \beta = -\frac{b}{a} = -\frac{-6}{2}$ $= 3$	$\alpha + \beta$ $-\frac{3}{2}$
$\alpha\beta = \frac{c}{a}$ $= \frac{1}{2}$	$\alpha\beta$ $\frac{5}{2}$
$\alpha^2 + \beta^2$ using $(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2$ $(\alpha + \beta)^2 - 2\alpha\beta = \alpha^2 + \beta^2$ $\alpha^2 + \beta^2 = (3)^2 - 2\left(\frac{1}{2}\right)$ $= 8$	$\alpha^2 + \beta^2$ $\frac{25}{4}$

ⓘ Sum of Squares Identity

This identity is helpful in solving problems involving sum and product of zeroes:

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

Example Apply sum and product of zeroes formulas for quadratics

Worked Example	Your Turn
Show that $x = 3$ is a zero of $P(x) = 2x^2 - 9x + 9$, Hence find the other zero. $P(3) = 2(3)^2 - 9(3) + 9 = 0$ Using $\alpha + \beta = -\frac{b}{a}$ $3 + \beta = \frac{9}{2}$ $\beta = \frac{3}{2}$	Show that $x = -2$ is a zero of $P(x) = 3x^2 + 5x - 2$, Hence find the other zero. $\frac{1}{3}$
Find p if one root of $x^2 + 2x - 5p = 0$ is double the other root. Let roots be α and 2α. Sum of roots: $\alpha + 2\alpha = -2$ $3\alpha = -2$ $\alpha = -\frac{2}{3}$ $\therefore$ roots are $-\frac{2}{3}$ and $-\frac{4}{3}$ Product of roots: $\alpha(2\alpha) = -5p$ $2\alpha^2 = -5p$ $p = -\frac{2\alpha^2}{5}$ $= \frac{-2\left(-\frac{2}{3}\right)^2}{5}$ $= -\frac{8}{45}$	Find q if one root of $x^2 - 4x + q = 0$ is triple the other root. $q = 3$

1 Let α and β are the roots of the given quadratic equation.
For each equation, find:

a $x^2 + 2x + 8 = 0$

$\alpha + \beta$: 2

$\alpha\beta$: 8

b $3x^2 - 6x - 2 = 0$

$\alpha + \beta$: -2

$\alpha\beta$: $-\frac{2}{3}$

c $4x^2 - 9x - 12 = 0$

$\alpha + \beta$: $\frac{9}{4}$

$\alpha\beta$: -3

d $5x^2 + 15x = 0$

$\alpha + \beta$: -3

$\alpha\beta$: 0

2 The equation of a monic quadratic polynomial with roots α and β is:

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Use this to find the quadratic equation, with integer coefficients, whose roots are:

a ± 5: $x^2 - 25 = 0$

b 3 and -5: $x^2 + 2x - 15 = 0$

c $2 \pm \sqrt{3}$: $x^2 - 4x + 1 = 0$

d $\frac{2}{5}$ and $\frac{5}{2}$: $10x^2 - 29x + 10 = 0$

3 Let the roots of $4x^2 - 5x - 1 = 0$ be α and β. Find the value of the following, without explicitly solving for α and β.

a $\alpha + \beta$: $\frac{5}{4}$

b $\alpha\beta$: $-\frac{1}{4}$

c $\frac{1}{\alpha} + \frac{1}{\beta}$: -5

d $(\alpha - 3)(\beta - 3)$: 5

e $\alpha^2 + \beta^2 + 2\alpha\beta$: $\frac{25}{16}$

f $\alpha^2\beta + \beta^2\alpha$: $-\frac{5}{16}$

4

a Expand $(\alpha + \beta)^2$

$\alpha^2 + 2\alpha\beta + \beta^2$

b Hence, express $\alpha^2 + \beta^2$ in terms of $(\alpha + \beta)$ and $\alpha\beta$.

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

c Let the roots of $x^2 - 3x - 2 = 0$ be α and β. Find the value of $\alpha^2 + \beta^2$

13

5 Let α and β be the roots of $x^2 - 4x + 2 = 0$. Find:

a $\alpha + \beta$ 4	**b** $\alpha\beta$ 2	**c** $\alpha^2\beta + \alpha\beta^2$ 8
d $\frac{1}{\alpha} + \frac{1}{\beta}$ 2	**e** $(\alpha + 2)(\beta + 2)$ 14	**f** $\alpha^2 + \beta^2$ 12
g $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ 6	**h** $\alpha\beta^3 + \alpha^3\beta$ 24	**i** $\left(\alpha + \frac{1}{\alpha}\right)\left(\beta + \frac{1}{\beta}\right)$ $\frac{17}{2}$

6 Suppose that the quadratic equation $x^2 - kx - 4x - 35 = 0$ has the sum of roots being 2. Find the value of k.

$k = -2$

7 If α and β are the zeroes of $2x^2 + 5x - 4 = 0$, find:

a $\alpha + \beta$

$-\frac{5}{2}$

b $\alpha\beta$

-2

c $\alpha^2 + \beta^2$

$\frac{41}{4}$

d $|\alpha - \beta|$
(hint: square it)

$\frac{\sqrt{57}}{2}$

8 One zero of $x^2 - 3x + k - 2 = 0$ is -4. Find the value of k.

$-4 + \beta = 3, \therefore \beta = 7$

$-4\beta = k - 2,$

$-28 = k - 2, \therefore k = -26$

Alternate method:

$P(-4) = 16 + 12 + k - 2 = 0$

$k = -26$

9 Find the value(s) of k if the quadratic equation $x^2 - (k+2)x + k + 1 = 0$ has:

a Equal roots

$k = 0$

b One root equal to 5

$k = 4$

c Consecutive roots

$k = \pm 1$

d One root double the other

$k = -\frac{1}{2}, 1$

e Reciprocal roots

$k = 0$

10 Let $3 - \sqrt{2}$ be a root of the quadratic equation $x^2 + mx + 7 = 0$, where m is a rational number. Find the

a other root

$3 + \sqrt{2}$

b value of m

-6

11

a Suppose the quadratic equation $4x^2 - px - 5 = 0$ has two roots that are of equal magnitude but opposite in sign. Find the value of p.

Let roots be $\alpha, -\alpha$

Sum of roots:$\alpha + (-\alpha) = -\frac{b}{a}$

$0 = -\frac{p}{4}$

$p = 0$

b Suppose the quadratic equation $x^2 + 5x + 3p - 1 = 0$ has two roots that are reciprocals of each other. Find the value of p.

Let roots be $\alpha, \frac{1}{\alpha}$

Product of roots: $\alpha \times \frac{1}{\alpha} = \frac{c}{a}$

$1 = 3p - 1$

$p = \frac{2}{3}$

12 If one of the roots of $x^2 + bx + c = 0$ is twice the other root, show that $2b^2 = 9c$.

Let roots be $\alpha, 2\alpha$

Sum of roots: $3\alpha = -b$ (1)

Product of roots: $2\alpha^2 = c$ (2)

Rearrange (1): $\alpha = -\frac{b}{3}$, Sub into (2)

$2\left(-\frac{b}{3}\right)^2 = c$

$\frac{2b^2}{9} = c$

$2b^2 = 9c$

13 If the roots of the equation $x^2 + px + q = 0$ are consecutive numbers, show that $p^2 = 4q + 1$.

Let roots be $\alpha, \alpha + 1$

Sum of roots: $2\alpha + 1 = -p$ (1)

Product of roots: $\alpha^2 + \alpha = q$ (2)

Rearrange (1): $\alpha = \frac{-p-1}{2}$, Sub into (2)

$\left(\frac{-p-1}{2}\right)^2 + \frac{-p-1}{2} = q$

$\frac{p^2 + 2p + 1}{4} + \frac{-2p-2}{4} = q$

$p^2 - 1 = 4q$

$p^2 = 4q + 1$

14 One root of $x^2 + px + q = 0$ is three times the other. Show that $3p^2 = 16q$.

Let roots be $\alpha, 3\alpha$

Sum of roots $4\alpha = -p$ (1)

Product of roots $3\alpha^2 = q$ (2)

Rearrange (1): $\alpha = -\frac{p}{4}$, Sub into (2)

$3\left(-\frac{p}{4}\right)^2 = q$

$\frac{3p^2}{16} = q$

$3p^2 = 16q$

15 Let the roots of $x^2 - 3x + 10 = 0$ be α and β.
Find the equation of the quadratic with roots $\alpha + 2$ and $\beta + 2$.

$\alpha + \beta = 3, \alpha\beta = 10$

Let new quadratic be $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

Sum of roots $(\alpha + 2) + (\beta + 2) = -\frac{b}{a}$

$\alpha + \beta + 2 = -\frac{b}{a}$

$7 = -\frac{b}{a}$

Product of roots: $(\alpha + 2)(\beta + 2) = \frac{c}{a}$

$\alpha\beta + 2(\alpha + \beta) + 4 = \frac{c}{a}$

$10 + 2(3) + 4 = \frac{c}{a}$

$20 = \frac{c}{a}$

$\therefore x^2 - 7x + 20 = 0$

16 If α and β are the roots of $5x^2 + mx - 5 = 0$, find the exact value of $(5\alpha + m)(5\beta + m)$

$\alpha + \beta = -\frac{m}{5}, \alpha\beta = -1$

$(5\alpha + m)(5\beta + m)$

$= 25\alpha\beta + 5\alpha m + 5\beta m + m^2$

$= 25\alpha\beta + 5m(\alpha + \beta) + m^2$

$= 25(-1) + 5m\left(-\frac{m}{5}\right) + m^2$

$= -25 - m^2 + m^2$

$= -25$

17 Let α and β be the roots of $x^2 - x + 4 = 0$.

a Find the value of $\alpha^2 + \beta^2$.

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= 1 - 2(4)$

$= -7$

b Hence explain why the roots of this quadratic are not real.

The sum of two squares is negative. This is not possible with real numbers.

18 Let α and β be the roots of $2x^2 - 8x + 1 = 0$. Find the quadratic equation whose roots are:

a $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

$\alpha + \beta = 4, \alpha\beta = \frac{1}{2}$

Let new quadratic be $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{a}$

$\frac{\beta + \alpha}{\alpha\beta} = -\frac{b}{a}$

$8 = -\frac{b}{a}$

$\frac{1}{\alpha\beta} = \frac{c}{a}$

$2 = \frac{c}{a}$

$x^2 - 8x + 2 = 0$

b α^2 and β^2

$\alpha + \beta = 4, \alpha\beta = \frac{1}{2}$

Let new quadratic be $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$\alpha^2 + \beta^2 = -\frac{b}{a}$

$(\alpha + \beta)^2 - 2\alpha\beta = -\frac{b}{a}$

$4^2 - 1 = -\frac{b}{a}$

$15 = -\frac{b}{a}$

$\alpha^2\beta^2 = \frac{c}{a}$

$\frac{1}{4} = \frac{c}{a}$

$x^2 - 15c + \frac{1}{4} = 0$

$\therefore 4x^2 - 60x + 1 = 0$

Sum and Product of Roots (Cubic)

Sum and Product of Zeroes for a Quadratic Equation

Let $ax^3 + bx^2 + cx + d = 0$, we can make this polynomial monic by $x^3 + \frac{b}{a}x^2 + \frac{c}{a}x + \frac{d}{a} = 0$ suppose it can be factorised to $P(x) = (x - \alpha)(x - \beta)(x - \gamma)$.

By expanding $P(x) = (x - \alpha)(x - \beta)(x - \gamma)$, equate coefficients to determine the relationship between zeroes and coefficients.

ⓘ Sum and Product of Zeroes (Quadratic)

For a **cubic** $P(x) = ax^3 + bx^2 + cx + d$ with zeroes α, β, γ:

$$\alpha + \beta = -\frac{b}{a}$$
$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$
$$\alpha\beta\gamma = -\frac{d}{a}$$

⚠ Describe $\alpha\beta + \alpha\gamma + \beta\gamma$ as 'sum of products taken two at a time'

ⓘ Sum of Squares Identity (3 zeroes)

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

Example Apply sum and product of zeroes formulas for cubics

α, β, γ are the roots of $2x^3 - 5x^2 + x - 1 = 0$, Find:

$(\alpha + \beta + \gamma)^2$

$$\alpha + \beta + \gamma = -\frac{b}{a} = \frac{5}{2}$$
$$(\alpha + \beta + \gamma)^2 = \left(\frac{5}{2}\right)^2$$
$$= \frac{25}{4}$$

$(\alpha + 1)(\beta + 1)(\gamma + 1)$

$$= (\alpha + 1)(\beta\gamma + \beta + \gamma + 1)$$
$$= \alpha\beta\gamma + \alpha\beta + \alpha\gamma + \alpha + \beta\gamma + \beta + \gamma + 1$$
$$= \alpha\beta\gamma + (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) + 1$$
$$= -\frac{d}{a} + \frac{c}{a} + \left(-\frac{b}{a}\right) + 1$$
$$= \frac{1}{2} + \frac{1}{2} + \frac{5}{2} + 1$$
$$= \frac{9}{2}$$

Guided Practice

α, β and γ be the roots of $x^3 - 3x + 2 = 0$. Find:

a $(\alpha + \beta + \gamma)^2$

b $(\alpha + 1)(\beta + 1)(\gamma + 1)$

c $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

d $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$

e $\alpha^2\beta^2\gamma + \alpha^2\gamma^2\beta + \beta^2\gamma^2\alpha$

f $\alpha^2 + \beta^2 + \gamma^2$

ⓘ Proofs involving Sum and Product of Roots

Harder questions require you to prove general statements about polynomials.

There is no step-by-step procedure for these questions, but it is helpful to:

- Clearly define the roots using the restrictions.
- Begin by writing out the sums and products of roots formulas.
- Use the sum equations and product equations first, as they are most simple.
- Use simultaneous equations to show the required statement.
- It is often easier if you do **not** find expressions for individual roots.

Guided Practice

a If one zero of the cubic $f(x) = ax^3 + bx^2 + cx + d$ is the opposite of the other, prove that $ad = bc$.

b If two of the roots of the equation $x^3 + qx + r = 0$ are equal, show that $4q^3 + 27r^2 = 0$.

1 Where α, β, γ are roots of the equation, for each cubic find:

a $x^3 + x^2 - 2x + 8 = 0$

$\alpha + \beta + \gamma$	$\alpha\beta + \alpha\gamma + \beta\gamma$	$\alpha\beta\gamma$
-1	-2	-8

b $2x^3 - x^2 + 6x + 2 = 0$

$\alpha + \beta + \gamma$	$\alpha\beta + \alpha\gamma + \beta\gamma$	$\alpha\beta\gamma$
$\frac{1}{2}$	3	-1

c $-x^3 - 3x^2 - 11 = 0$

$\alpha + \beta + \gamma$	$\alpha\beta + \alpha\gamma + \beta\gamma$	$\alpha\beta\gamma$
-3	0	-11

d $x^3 + 7x - 3 = 0$

$\alpha + \beta + \gamma$	$\alpha\beta + \alpha\gamma + \beta\gamma$	$\alpha\beta\gamma$
0	7	3

2 If α, β, and γ are the roots of $2x^3 + 5x^2 - x - 3 = 0$, find:

a $\alpha\beta\gamma$ — $\frac{3}{2}$

b $\alpha\beta + \alpha\gamma + \beta\gamma$ — $-\frac{1}{2}$

c $\alpha + \beta + \gamma$ — $-\frac{5}{2}$

d $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ — $-\frac{1}{3}$

e $(\alpha + 1)(\beta + 1)(\gamma + 1)$ — $-\frac{1}{2}$

3 If α, β, and γ are the roots of $x^3 + 2x^2 - 11x - 12 = 0$, find:

a $\alpha + \beta + \gamma$ — -2

b $\alpha\beta + \alpha\gamma + \beta\gamma$ — -11

c $\alpha\beta\gamma$ — 12

d $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ — $-\frac{11}{12}$

e $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma}$ — $-\frac{1}{6}$

f $(\alpha + 1)(\beta + 1)(\gamma + 1)$ — 0

g $\alpha^2\beta^2\gamma + \alpha^2\gamma^2\beta + \beta^2\gamma^2\alpha$ — -132

h $\alpha^2 + \beta^2 + \gamma^2$ — 26

i $(\alpha\beta)^{-2} + (\alpha\gamma)^{-2} + (\beta\gamma)^{-2}$ — $\frac{13}{72}$

4 If two roots of the cubic equation $x^3 + ax^2 + bx + 24 = 0$ are 4 and -2.
Find the values of a and b and hence state equation of the cubic.

Product of roots: $(4)(-2)\gamma = -24$

$\gamma = 3$

Sum of roots: $4 - 2 + 3 = -a;\ a = -5$

Sum of products: $-8 + 12 - 6 = b;\ b = -2$

$\therefore x^3 - 5x^2 - 2x + 24 = 0$

5

a The polynomial $P(x) = 2x^3 - 5x^2 - 14x + 8$ has zeroes at -2 and 4.
Use the sum of zeroes to find the other zero.

Sum of roots: $-2 + 4 + \gamma = \frac{5}{2}$

$\gamma = \frac{1}{2}$

b Suppose that $x - 3$ and $x + 1$ are factors of $P(x) = x^3 - 6x^2 + 5x + 12$, use the product of the zeroes to find the other factor of $P(x)$.

Two roots are 3 and -1.

Product of roots: $(3)(-1)\gamma = -12$

$\gamma = 4$

The other factor is $x - 4$

6 Consider $P(x) = x^3 - x^2 - x + 10$.

a Show that -2 is a zero of $P(x)$.

$P(-2) = (-2)^3 - (-2)^2 - (-2) + 10 = 0$

b Suppose that the zeroes of $P(x)$ are $-2, \alpha$, and β. Show that $\alpha + \beta = 3$ and $\alpha\beta = 5$.

$\alpha + \beta - 2 = 1, \quad \therefore\ \alpha + \beta = 3$

$-2\alpha\beta = -10, \quad \therefore\ \alpha\beta = 5$

c By solving the two equations in part **b** simultaneously, show that $\alpha^2 - 3\alpha + 5 = 0$

$\beta = 3 - \alpha$, sub into $\alpha\beta = 5$

$\alpha(3 - \alpha) = 5$

$3\alpha - \alpha^2 = 5$

$\therefore \alpha^2 - 3\alpha + 5 = 0$

d Hence show there are no such real numbers α and β.

$\Delta = 9 - 4(1)(5) < 0$

So α is not real, hence β is not real either.

e Hence state how many times the graph of $P(x)$ crosses the x-axis.

Only once.

7 Two of the roots of $x^3 - 2x^2 - 9x + 18 = 0$ sum to zero.

a Since two of the roots sum to zero, we can let the roots be ……, ……, and β.

$\alpha, -\alpha$

b Use the sum of roots to show that $\beta = 2$.

$\alpha + (-\alpha) + \beta = -(-2)$

$\therefore \beta = 2$

c If one of the roots is 2, write the corresponding linear factor.

$x - 2$

d Using long division, or otherwise, completely factorise $x^3 - 2x^2 - 9x + 18 = 0$

By long division, $P(x) = (x - 2)(x^2 - 9)$

e Hence, find all the roots of $x^3 - 2x^2 - 9x + 18 = 0$.

$-3, 2, 3$

8 Show that $x = 1$ and $x = -2$ are zeroes of $P(x)$, and use the sum or product of zeroes of find the other zero.

a $P(x) = x^3 - 2x^2 - 5x + 6$

$P(1) = 0$

$P(-2) = 0$

Sum of roots: $1 - 2 + \gamma = 2$

$\gamma = 3$

b $P(x) = 2x^3 + 3x^2 - 3x - 2$

$P(1) = 0$

$P(-2) = 0$

Sum of roots: $1 - 2 + \gamma = -\frac{3}{2}$

$\gamma = -\frac{1}{2}$

9 Two of the roots of $x^3 - px^2 - qx + 30 = 0$ are 3 and 5.

a Find the other root.

Product of roots: $(3)(5)\gamma = -30$

$\therefore \gamma = -2$

b Find p and q.

Sum of roots: $3 + 5 + (-2) = p;\ \therefore p = 6$

Sum of products: $3(5) + 3(-2) + 5(-2) = -q;\ \therefore q = 1$

$\therefore x^3 - 6x^2 - x + 30 = 0$

10 Find the roots of $x^3 - 3x^2 + 4 = 0$ given that 2 of the roots are equal.

Let the roots be α, α, β

Sum of roots: $\alpha + \alpha + \beta = 3$

$\therefore\ 2\alpha + \beta = 3 \ldots (1)$

Product of roots: $\alpha^2\beta = -4$ ... (2)

Sum of products: $\alpha^2 + \alpha\beta + \alpha\beta = 0$

$\therefore\ \alpha(\alpha + 2\beta) = 0$

Since $\alpha \neq 0$ (otherwise product would be 0),

$\alpha + 2\beta = 0$

$\therefore\ \beta = -\frac{\alpha}{2} \ldots (3)$

Substituting (3) into (1):

$2\alpha - \frac{\alpha}{2} = 3$

$\frac{3\alpha}{2} = 3$

$\therefore\ \alpha = 2$

From (3): β = -2/2 = -1

$\therefore x = -1, 2, 2$

11 Solve $12x^3 - 4x^2 - 3x + 1 = 0$ if the sum of 2 of its roots is 0.

Let roots be $\alpha, -\alpha, \beta$

Sum of roots: $\beta = \frac{1}{3}$

Product of roots: $-\alpha^2\beta = -\frac{1}{12}$

$-\alpha^2\left(\frac{1}{3}\right) = -\frac{1}{12}$

$\alpha^2 = \frac{1}{4}$

$\alpha = \pm\frac{1}{2}$

$x = \frac{1}{3}, \pm\frac{1}{2}$

12 Two roots of $x^3 + mx^2 - 3x - 18 = 0$ are equal and rational. Find m.

Let roots be α, α, β

sum: $2\alpha + \beta = -m \ldots (1)$

product: $\alpha^2\beta = 18 \ldots (2)$

sum of products: $\alpha^2 + 2\alpha\beta = -3 \ldots (3)$

from (2), $\beta = \frac{18}{\alpha^2}$ sub into (3)

$\alpha^2 + 2\alpha\left(\frac{18}{\alpha^2}\right) = -3$

$\therefore\ \alpha^3 + 3\alpha + 36 = 0$

Test factors of 36 to find α: $P(-3) = 0$

$\therefore\ \alpha = -3$

By long division, $\alpha^3 + 3\alpha + 36 = (\alpha + 3)(\alpha^2 - 3\alpha + 12)$

$\Delta < 0$ for $\alpha^2 - 3\alpha + 12$

so $\alpha = -3$ is the only rational root.

$\therefore\ \beta = \frac{18}{(-3)^2} = 2$

Sum of roots $-3 - 3 + 2 = -m$

$\therefore\ m = 4$

13 A set of numbers is an **arithmetic progression** if the difference between successive terms is constant. For example, {1, 4, 7} is an arithmetic progression with common difference 3.
The polynomial equation $x^3 - 3x^2 - x + 3 = 0$ has its roots in arithmetic progression.

a Let the roots be $(\alpha - d), \alpha,$ and $(\alpha + d)$, where d is the common difference.
Use sum of roots to find a root.

sum of roots: $3\alpha = 3$

$\therefore \alpha = 1$

b Write down the corresponding linear factor, then completely factorise $x^3 - 3x^2 - x + 3 = 0$.

If $\alpha = 1$ is a root, then $x - 1$ is the linear factor.

by long division: $(x - 1)(x^2 - 2x - 3)$

c Therefore, find the roots of $x^3 - 3x^2 - x + 3 = 0$.

$(x - 1)(x^2 - 2x - 3) = (x - 1)(x - 3)(x + 2)$

$\therefore x = -1, 1, 3$

14 A set of numbers is a **geometric progression** if the ratio between successive terms is constant. For example, {2, 6, 18} is a geometric progression with common ratio 3.
The polynomial equation $x^3 - 7x^2 + 14x - 8 = 0$ has its roots in geometric progression.

a Let the roots be $\frac{\alpha}{r}, \alpha,$ and αr, where r is the common ratio.
Use product of roots to find a root.

product of roots: $\alpha^3 = 8$

$\therefore \alpha = 2$

b Therefore, completely factorise $x^3 - 7x^2 + 14x - 8 = 0$.

By long division: $(x - 2)(x^2 - 5x + 4)$

c Therefore, find the roots of $x^3 - 7x^2 + 14x - 8 = 0$.

$(x - 2)(x^2 - 5x + 4) = (x - 2)(x - 4)(x - 1)$

$\therefore x = 1, 2, 4$

15 Find the roots of these polynomial equations, given that

a The roots of $x^3 - 6x^2 + 3x + 10 = 0$ are in arithmetic progression.

Let roots be $(\alpha - d), \alpha, (\alpha + d)$

Sum of roots: $3\alpha = 6; \therefore \alpha = 2$

Product of roots: $(2 - d)(2)(2 + d) = -10$

$\therefore 8 - 2d^2 = -10$

$\therefore d = \pm 3$

$\therefore x = -1, 2, 5$

You can also solve this using long division, as above.

b The roots of $3x^3 - 14x^2 - 28x + 24 = 0$ are in geometric progression.

Let the roots be $\frac{\alpha}{r}, \alpha, \alpha r$

Sum of roots: $\frac{\alpha}{r} + \alpha + \alpha r = \alpha\left(\frac{1}{r} + 1 + r\right) = \frac{14}{3} \ldots (1)$

Product of roots: $\left(\frac{\alpha}{r}\right)(\alpha)(\alpha r) = \alpha^3 = -8; \therefore \alpha = -2$

Substituting $\alpha = -2$ into (1): $-2\left(\frac{1}{r} + 1 + r\right) = \frac{14}{3}$

This simplifies to $3r^2 + 10r + 3 = 0$

$r = -3$ or $-\frac{1}{3}$

$\therefore x = \frac{2}{3}, -2, 6$

16 Two of the roots of $ax^3 + bx^2 + cx + d = 0$ are equal in magnitude but opposite in sign.

a Show that one of the roots is $x = -\frac{b}{a}$.

Let roots be $\alpha, -\alpha, \beta$

Sum of roots: $\alpha + (-\alpha) + \beta = -\frac{b}{a}$

$\therefore \beta = -\frac{b}{a}$

b By substituting this back into the polynomial, show that $ad = bc$.

$a\left(-\frac{b}{a}\right)^3 + b\left(-\frac{b}{a}\right)^2 + c\left(-\frac{b}{a}\right) + d = 0$

$-\frac{b^3}{a^2} + \frac{b^3}{a^2} - \frac{bc}{a} + d = 0$

$-\frac{bc}{a} + d = 0$

$-bc + ad = 0$

$bc = ad$

17 Find the values of a and b for which $x^3 + ax^2 - 10x + b$ is exactly divisible by $x^2 + x - 12$, and then factor the cubic.

$x^2 + x - 12 = (x - 3)(x + 4)$

$P(3) = 0;\ 27 + 9a - 30 + b = 0$

$\therefore 9a + b = 3$

$P(-4) = 0;\ -64 + 16a + 40 + b = 0$

$16a + b = 24$

solving simultaneously, $a = 3, b = -24$

$\therefore x^3 + 3x^2 - 10x - 24$

From here, you can either use long division or sum of roots to find the third root

$(x - 3)(x + 4)(x + 2)$

18 Find the roots of $3x^3 - x^2 - 48x + 16 = 0$ given the sum of two of them is zero.

Hint: let the roots be $\alpha, -\alpha, \beta$

Let the roots be α, -α, β

Sum of roots: $\beta = \frac{1}{3}$

Product of roots: $-\alpha^2\beta = -\left(\frac{16}{3}\right)$

$-\alpha^2\left(\frac{1}{3}\right) = -\left(\frac{16}{3}\right)$

$\therefore \alpha = \pm 4$

$\therefore x = \frac{1}{3}, -4, 4$

19 Find the roots of $2x^3 - 5x^2 - 46x + 24 = 0$ given that the product of two of them is 3.

Hint: let the roots be $\alpha, \frac{3}{\alpha}, \beta$

Let the roots be $\alpha, \frac{3}{\alpha}, \beta$

Sum of roots: $\alpha + \left(\frac{3}{\alpha}\right) + \beta = \frac{5}{2} \ldots (1)$

Product of roots: $\alpha\left(\frac{3}{\alpha}\right)\beta = -12$

$\therefore \beta = -4$

Substituting $\beta = -4$ into (1):

$\alpha + \left(\frac{3}{\alpha}\right) + (-4) = \frac{5}{2}$

$\alpha + \left(\frac{3}{\alpha}\right) = \frac{13}{2}$

this rearranges to $2\alpha^2 - 13\alpha + 6 = 0$

$(2\alpha - 1)(\alpha - 6) = 0$

If $\alpha = 6$, $\frac{3}{\alpha} = \frac{1}{2}$, if $\alpha = \frac{1}{2}$, $\frac{3}{\alpha} = 6$

$\therefore x = 6, \frac{1}{2}, -4$

20 Find the roots of $x^3 - 27x - 54 = 0$ given that two of them are equal.

Hint: let the roots be α, α, β

Sum of roots: $\alpha + \alpha + \beta = 0$

$\therefore \beta = -2\alpha \ldots (1)$

Product of roots: $\alpha^2\beta = 54 \ldots (2)$

Substituting (1) into (2):

$-2\alpha^3 = 54$

$\therefore \alpha = -3.\ \therefore \beta = 6$

$x = -3, -3, 6$

21 Find the zeroes of the polynomial $P(x) = 2x^3 - 13x^2 + 22x - 8$ given that one of them is the product of the other two.

Hint: let the roots be $\alpha, \beta, \alpha\beta$

sum of roots: $\alpha + \beta + \alpha\beta = \frac{13}{2}$

sum of products: $\alpha\beta + \alpha^2\beta + \alpha\beta^2 = 11$

$\alpha\beta(1 + \alpha + \beta) = 11$

product of roots: $\alpha^2\beta^2 = 4$

$\alpha\beta = \pm 2$

Consider $\alpha\beta = 2$

using sum of roots $\alpha + \beta + 2 = \frac{13}{2}$

$\alpha + \beta = \frac{9}{2}$

Check this satisfies $\alpha\beta(1 + \alpha + \beta) = 11$

$2\left(1 + \frac{9}{2}\right) = 11$ ✓

$\alpha\beta = -2$ does not satisfy sum of products, so reject it.

$\therefore \alpha + \beta = \frac{9}{2}, \quad \alpha\beta = 2$

Solve simultaneously to get $\alpha = 4, \frac{1}{2}$

$\therefore$ roots are $4, \frac{1}{2}, 2$

22

a Let α, β, γ be the roots of $2x^3 - 4x^2 + 3x - 1 = 0$.
Find the cubic equation whose roots are $2\alpha, 2\beta$, and 2γ.

Sum of roots: $\alpha + \beta + \gamma = 2,$

Sum of products: $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{3}{2}$

Product of roots: $\alpha\beta\gamma = \frac{1}{2}$

For new roots 2α, 2β, 2γ:

Sum of new roots: $2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = 2(2) = 4$

Sum of products of new roots: $(2\alpha)(2\beta) + (2\alpha)(2\gamma) + (2\beta)(2\gamma) = 4\alpha\beta + 4\alpha\gamma + 4\beta\gamma = 4(\alpha\beta + \alpha\gamma + \beta\gamma) = 4\left(\frac{3}{2}\right) = 6$

Product of new roots: $(2\alpha)(2\beta)(2\gamma) = 8\alpha\beta\gamma = 8\left(\frac{1}{2}\right) = 4$

Let the new cubic equation be $x^3 - \frac{b}{a}x + \frac{c}{a}x - \frac{d}{a} = 0$

From sums and products of new roots: $x^3 - 4x^2 + 6x - 4 = 0$

b Let $x^3 - 3x^2 + 5x + 2 = 0$ have zeroes α, β, and γ.
Find the equation of the polynomial equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$.

Sum of roots: $\alpha + \beta + \gamma = 3,$ Sum of products: $\alpha\beta + \alpha\gamma + \beta\gamma = 5,$ Product of roots: $\alpha\beta\gamma = -2$

For new roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$:

Sum of new roots: $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{5}{-2} = -\frac{5}{2}$

Sum of products of new roots: $\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right) + \left(\frac{1}{\alpha}\right)\left(\frac{1}{\gamma}\right) + \left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right) = \frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = \frac{\gamma + \beta + \alpha}{\alpha\beta\gamma} = \frac{3}{-2} = -\frac{3}{2}$

Product of new roots: $\left(\frac{1}{\alpha}\right)\left(\frac{1}{\beta}\right)\left(\frac{1}{\gamma}\right) = \frac{1}{\alpha\beta\gamma} = \frac{1}{-2} = -\frac{1}{2}$

$\therefore 2x^3 + 5x^2 - 3x + 1 = 0$

23 Let $x^3 + 5x - 1 = 0$ have roots α, β and γ.

a Find the value of $\alpha^2 + \beta^2 + \gamma^2$.

$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$= 0^2 - 2(5)$

-10

b Hence, explain why the graph of $y = x^3 + 5x - 1$ cannot have 3 x-intercepts.

A cubic has 3 roots.

The sum of the squares of roots is negative, i.e., at least 1 non-real root, so cannot have 3 x-intercepts.

24 Two of the roots of $x^3 + 3x^2 - 4x + a = 0$ are opposites.
Find the value of a and the three roots.

Let roots be $\alpha, -\alpha, \beta$

Sum of roots: $\beta = -3$

Sum of product of pairs of roots: $-\alpha^2 + \alpha\beta - \alpha\beta = -4$

$\alpha^2 = 4$

$\alpha = \pm 2$

$a = -12$ and the roots are −2, 2 and −3

25 Two of the roots of $4x^3 + ax^2 - 47x + 12 = 0$ are reciprocals.
Find the value of a and the three roots.

Let roots be $\alpha, \frac{1}{\alpha}, \beta$

Product of roots: $\beta = -3$

Sum of roots: $\alpha + \frac{1}{\alpha} + \beta = -\frac{a}{4}$

$\alpha + \frac{1}{\alpha} - 3 = -\frac{a}{4}$

$\alpha + \frac{1}{\alpha} = 3 - \frac{a}{4}$ (1)

Sum of product of pairs of roots: $1 - \frac{3}{\alpha} - 3\alpha = -\frac{47}{4}$

$1 - 3\left(\alpha + \frac{1}{\alpha}\right) = -\frac{47}{4}$

$\alpha + \frac{1}{\alpha} = \frac{17}{4}$ (2)

$4\alpha^2 - 17\alpha + 4 = 0$

$(4\alpha - 1)(\alpha - 4) = 0$

$\alpha = \frac{1}{4}, 4$

Sub (1) into (2)

$\frac{17}{4} = 3 - \frac{a}{4}$

$\therefore a = -5$

$a = -5$ and the roots are $4, \frac{1}{4}$ and -3.

26 2019 HSC Extension 1 Band 4

$P(x) = qx^3 + rx^2 + rx + q$, where q and r are constants and $q \neq 0$.

One of the zeroes of $P(x)$ is -1, given that α is a zero of $P(x)$, which of the following is also a zero?

A. $-\frac{1}{\alpha}$

B. $-\frac{q}{\alpha}$

C. $\frac{1}{\alpha}$

D. $\frac{q}{\alpha}$

Let roots be $\alpha, \beta, -1$

$-\alpha\beta = -\frac{q}{q} = -1$

$\alpha\beta = 1$

$\therefore \beta = \frac{1}{\alpha}$

C

27 HSC Sample Question Extension 1 Band 5

The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots $\sqrt{k}, -\sqrt{k}$, and α.

a Show that $\alpha + p = 0$.

Using sum of roots $\sqrt{k} - \sqrt{k} = -\frac{b}{a}$

$a = -p$

$\alpha + p = 0$

b Show that $k\alpha = r$.

Using product of roots: $\alpha\sqrt{k}(-\sqrt{k}) = -\frac{d}{a}$

$-k\alpha = -r$

$k\alpha = r$

c Show that $pq = r$.

Using product of pairs of roots: $\sqrt{k}(-\sqrt{k}) + \alpha\sqrt{k} - \alpha\sqrt{k} = \frac{c}{d}$

$-k = q$

Substitute into part **b**

$-q\alpha = r$

Using $-\alpha = p$ from part **a**

$qp = r$

28 2006 HSC Extension 1 Band 5

$P(x) = x^3 + rx^2 + sx + t$ has three real zeroes $1, \alpha$, and $-\alpha$

a Find the value of r.

Use sum of roots. $r = -1$

b Find the value of $s + t$.

Using $r = -1$ and $P(1) = 0$

$1 - 1 + s + t = 0$

$\therefore s + t = 0$

29 If one root of the equation $x^3 - bx^2 + cx - d = 0$ is equal to the product of the other two, show that $(c + d)^2 = d(b + a)^2$, where $a = 1$.

Let zeroes be $\alpha, \beta, \alpha\beta$

Sum of roots: $\alpha + \beta + \alpha\beta = b$

Sum of product of pairs of roots: $\alpha\beta + \alpha\beta^2 + \alpha^2\beta = c$

$$\alpha\beta(1 + \beta + \alpha) = c$$

Product of roots: $\alpha^2\beta^2 = d$

LHS = $(c + d)^2$

$= c^2 + 2cd + d^2$

$= \alpha^2\beta^2(1 + \beta + \alpha)^2 + 2\alpha^3\beta^3(1 + \beta + \alpha) + \alpha^4\beta^4$

$= \alpha^2\beta^2[(1 + \beta + \alpha)^2 + 2\alpha\beta(1 + \beta + \alpha) + \alpha^2\beta^2]$

recognise perfect square

$= \alpha^2\beta^2\big((1 + \alpha + \beta) + \alpha\beta\big)^2$

$= d(a + b)^2$ as required.

30 The cubic equation $x^3 - Ax^2 + 3A = 0$, where $A > 0$, has roots α, β, and $\alpha + \beta$.
Show that $A = 2\sqrt{6}$.

Sum of roots: $2\alpha + 2\beta = A$ (1)

Sum of product of pairs of roots: $\alpha\beta + \beta(\alpha + \beta) + \alpha(\alpha + \beta) = 0$ (2)

Product of roots: $\alpha\beta(\alpha + \beta) = -3A$ (3)

Rearrange (1): $\alpha + \beta = \frac{A}{2}$, sub into (2)

$$\alpha\beta + \beta\left(\frac{A}{2}\right) + \alpha\left(\frac{A}{2}\right) = 0$$

$$\alpha\beta + \frac{A}{2}(\alpha + \beta) = 0$$

$$\alpha\beta + \frac{A^2}{4} = 0$$

$$\alpha\beta = -\frac{A^2}{4}$$

Sub value of $\alpha + \beta$ and $\alpha\beta$ into (3)

$$-\frac{A^2}{4}\left(\frac{A}{2}\right) = -3A$$

$$-\frac{A^3}{8} = -3A$$

$$A^2 = 24$$

$$A = \sqrt{24}$$

$$A = 2\sqrt{6}$$

Sum and Product of Roots (Quartic)

ⓘ Sums and Products of Zeroes (Quartic)

For a **quartic** $P(x) = ax^4 + bx^3 + cx^2 + dx + e$:

$$\alpha + \beta + \gamma + \delta = -\frac{b}{a}$$
$$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a}$$
$$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -\frac{d}{a}$$
$$\alpha\beta\gamma\delta = \frac{e}{a}$$

All methods in the previous sections apply.

FOUNDATION

1 For each quartic equation, where $\alpha, \beta, \gamma, \delta$ are roots of the equation, find:

a $x^4 + 2x^3 - x^2 - x + 5 = 0$

$\alpha + \beta + \gamma + \delta$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ −2

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$ −1

$\alpha\beta\gamma\delta$ 1

5

b $2x^4 - 2x^3 - 4x^2 + 3x - 2 = 0$

$\alpha + \beta + \gamma + \delta$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ 1

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$ −2

$\alpha\beta\gamma\delta$ $-\frac{3}{2}$

−1

c $2x^4 - 12x^3 + 7 = 0$

$\alpha + \beta + \gamma + \delta$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$ 6

$\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$ 0

$\alpha\beta\gamma\delta$ 0

$\frac{7}{2}$

2 If $\alpha, \beta, \gamma, \delta$ are the roots of $x^4 - 2x^3 + 5x - 3 = 0$, find:

a $\alpha\beta\gamma\delta$

−3

b $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

−5

c $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$

$\frac{5}{3}$

3 If α, β, γ, and δ are the roots of $x^4 - 5x^3 + 2x^2 - 4x - 3 = 0$, find:

a $\alpha + \beta + \gamma + \delta$

5

b $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$

2

c $\alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta$

4

d $\alpha\beta\gamma\delta$

-3

d $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$

$-\frac{4}{3}$

e $\frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\alpha\delta} + \frac{1}{\beta\gamma} + \frac{1}{\beta\delta} + \frac{1}{\gamma\delta}$

$-\frac{2}{3}$

f $\frac{1}{\alpha\beta\gamma} + \frac{1}{\alpha\beta\delta} + \frac{1}{\alpha\gamma\delta} + \frac{1}{\beta\gamma\delta}$

$-\frac{5}{3}$

e $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

21

4

a Show that 1 is a zero of the polynomial $P(x) = x^4 - 2x^3 + 7x - 6$.

$P(1) = 1 - 2 + 7 - 6 = 0$

b If α, β, γ are the 3 other zeroes, find the value of $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$

$\alpha + \beta + \gamma = 1, \alpha\beta\gamma = -6$

5 The polynomial equation $x^4 + ax^3 + bx^2 + cx + d = 0$ has roots $\alpha, \beta, \gamma, \delta$.

a Show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = a^2 - 2b$.

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = (\alpha + \beta + \gamma + \delta)^2 - 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta)$

$= (-a)^2 - 2b = a^2 - 2b$ as required

b Hence, show that $x^4 - 3x^3 + 5x^2 + 4x - 1 = 0$ cannot have four real roots.

$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = a^2 - 2b = 9 - 10 = -1 < 0$

Since the sum of squares of roots is negative, there is at least 1 non-real root.

Since a quartic has 4 roots, not all 4 roots are real.

6 The polynomial equation $x^4 + px^3 + qx^2 + rx + s = 0$ has two roots that sum to zero, and another two roots being reciprocals of each other. Show that:

a $q = 1 + s$

$\alpha, -\alpha, \beta, \frac{1}{\beta}$ are the roots.

Product of roots: $-\alpha^2 = s$

Sum of product of pairs of roots: $-\alpha^2 + \alpha\beta + \frac{\alpha}{\beta} - \alpha\beta - \frac{\alpha}{\beta} + 1 = q$

$\therefore -\alpha^2 + 1 = q$

$\therefore s + 1 = q$

b $r = ps$

Sum of roots: $\beta + \frac{1}{\beta} = -p$

Sum of product of 3 roots: $-\alpha^2\beta - \frac{\alpha^2}{\beta} - \alpha + \alpha = r$

$-\alpha^2\beta - \frac{\alpha^2}{\beta} = r$

$-\alpha^2\left(\beta + \frac{1}{\beta}\right) = r$

$-s(-p) = r$

$sp = r$ as required.

7 If $x = 2$ is a double root of $ax^4 - 2x^3 - 8x + 16 = 0$, find the value of a and the sum of the other two roots.

$P(2) = 16a - 16 - 16 + 16 = 0$

$\therefore a = 1$

Let roots be $2, 2, \alpha, \beta$

Sum of roots $4 + \alpha + \beta = \frac{2}{a}$

$4 + \alpha + \beta = 2$

$\therefore \alpha + \beta = -2$

8 Show that $x = 1$ and $x = -2$ are zeroes of $P(x)$, then find the other two zeroes using sum and product of roots.

a $x^4 + 3x^3 - 3x^2 - 7x + 6$

$P(1) = 3 - 5 - 10 + 20 - 8 = 0$

$P(-2) = 3(-2)^4 - 5(-2)^3 - 10(-2)^2 + 20(-2) - 8 = 0$

Sum of roots: $\alpha + \beta - 2 + 1 = -3$

$\alpha + \beta = -2$

Product of roots: $-2\alpha\beta = 6$

$\alpha\beta = -3$

Solving simultaneously,

$\alpha(-2 - \alpha) = -3$

$\alpha^2 + 2\alpha - 3 = 0$

$(\alpha + 3)(\alpha - 1) = 0$

$\alpha = -3, 1$

If $\alpha = -3, \beta = 1$, if $\alpha = 1, \beta = -3$ (these are the same)

$\therefore$ other zeroes are −3, 1, so 1 is a double root.

b $P(x) = 3x^4 - 5x^3 - 10x^2 + 20x - 8$

$P(1) = 1 + 3 - 3 - 7 + 6 = 0$

$P(-2) = (-2)^4 + 3(-2)^3 - 3(-2)^2 - 7(-2) + 6 = 0$

Sum of roots: $\alpha + \beta - 2 + 1 = \frac{5}{3}$

$\alpha + \beta = \frac{8}{3}$

Product of roots: $-2\alpha\beta = -\frac{8}{3}$

$\alpha\beta = \frac{4}{3}$

Solving simultaneously,

$\alpha\left(\frac{8}{3} - \alpha\right) = \frac{4}{3}$

$3\alpha^2 - 8\alpha + 4 = 0$

$(3\alpha - 2)(\alpha - 2) = 0$

$\alpha = \frac{2}{3}, 2$

If $\alpha = \frac{2}{3}, \beta = \frac{6}{3} = 2$, if $\alpha = 2, \beta = \frac{2}{3}$ (these are the same)

$\therefore$ other zeroes are $\frac{2}{3}, 2$

9 Find the zeroes of $x^4 - 3x^3 - 8x^2 + 12x + 16$ if they are $\alpha, 2\alpha, \beta,$ and 2β.

Sum of roots: $3\alpha + 3\beta = 3$

$\alpha + \beta = 1$

Product of roots: $4\alpha^2\beta^2 = 16$

$\alpha^2\beta^2 = 4$

Solving simultaneously,

$\alpha^2(1-\alpha)^2 = 4$

$\alpha(1-\alpha) = \pm 2$

Consider 2 cases:

$\alpha(1-\alpha) = 2$	$\alpha(1-\alpha) = -2$
$\alpha^2 - \alpha + 2 = 0$	$\alpha^2 - \alpha - 2 = 0$
No solution as $\Delta < 0$	$(\alpha + 1)(\alpha - 2) = 0$
	$\therefore \alpha = -1, 2$

If $\alpha = -1$, $\beta = 2$, If $\alpha = 2, \beta = -1$, (these are the same)

$\therefore$ −1, −2, 2 and 4

10 The polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ has two zeroes that are opposites and two zeroes that are reciprocals. Show that $b = 1 + d$ and $c = ad$.

Let roots be $\alpha, -\alpha, \beta, \frac{1}{\beta}$

Sum of roots: $\beta + \frac{1}{\beta} = -a$ (1)

Sum of products 2 at a time: $-\alpha^2 + \alpha\beta + \frac{\alpha}{\beta} - \alpha\beta - \frac{\alpha}{\beta} + 1 = b$

$\therefore -\alpha^2 + 1 = b$ (2)

Sum of products 3 at a time: $-\alpha^2\beta - \frac{\alpha^2}{\beta} + r - r = -c$

$\therefore -\alpha^2\left(\beta - \frac{1}{\beta}\right) = -c$ (3)

Product of roots: $-\alpha^2 = d$ (4)

sub (4) into (2) $\rightarrow d + 1 = b$

sub (1) and (4) into (3) $\rightarrow d(-a) = -c$

$-ad = -c$

$c = ad$

Applications of Polynomials to Geometry Problems

ⓘ Applications of Polynomials to Geometry Problems

Find the points where two curves meet by setting their equations equal and solving for x.

- Each zero represents a point where curves intersect.
- No zeroes mean no intersections exist.
- Single roots show where curves pass through each other.
- Double roots indicate curves are tangent at that point (they touch and bounce away).

Sketch using DESMOS to visualise what's happening.

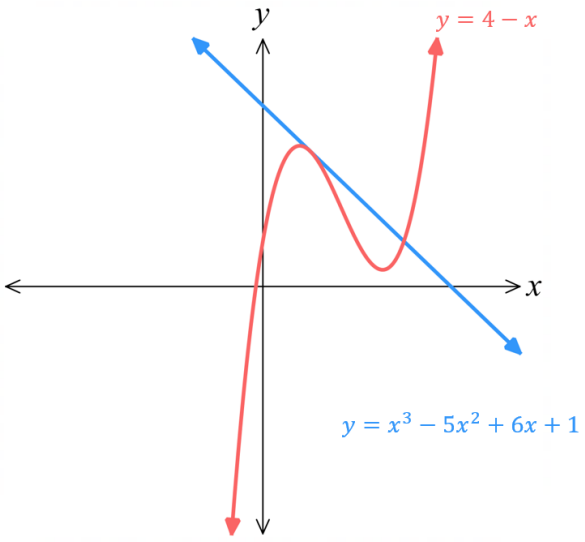
Guided Practice

Find the points of intersection of a cubic and a line.
Consider $y = 4 - x$ and $y = x^3 - 5x^2 + 6x + 1$.

a Equate the two curves to find an equation that can be used to find the points of intersection.

b Find two roots using factor theorem, then use sum of roots to find the third root.

c Hence explain why the line is tangent to the cubic, and state the points of intersection.



ⓘ Horizontal Distance between Zeroes

To find horizontal distance $|\alpha - \beta|$ between two intersection points, use this identity:

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta}$$

Guided Practice

The line $y = 2x - 2$ meets the cubic $y = x^3 - 5x^2 + 4x$ at $F(1,0)$ and also at points A and B.

a Show that the x coordinates of α and β satisfy $x^3 + 5x^2 + 2x + 2 = 0$.

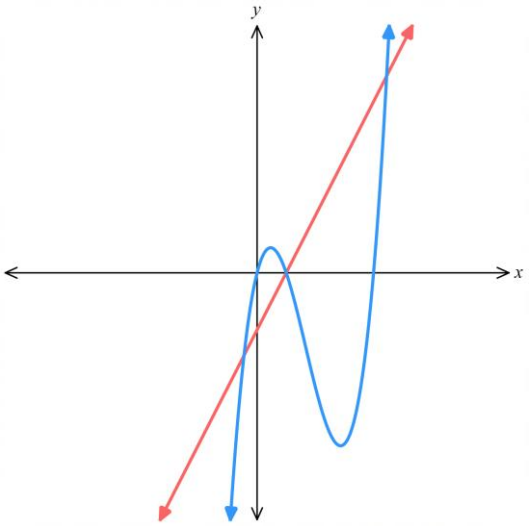
b Find $\alpha + \beta$.

c Hence find the coordinates of the midpoint M of AB.

d Show that $\alpha\beta = -2$.

e Use the identity $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ to show that horizontal distance $|\alpha - \beta|$ between A and B is $2\sqrt{6}$.

f Hence find the exact length of AB using the horizontal distance and the gradient.



1 Consider the parabola $y = x^2 - 6x$ and the line $y = 2x - 16$.

a Show that the x-coordinates of the points of intersection satisfy the equation $x^2 - 8x + 16 = 0$.

$x^2 - 6x = 2x - 16$

$\therefore x^2 - 8x + 16 = 0$

b By factorising the LHS of the equation, show that the line is tangent to the parabola, and state hence find the tangent point.

$x^2 - 8x + 16 = 0$

$(x - 4)^2 = 0$

4 is a double root, so the line is tangent to the parabola at $x = 4$.

Tangent point is $(4, -8)$

c Use DESMOS to visualise what is happening by sketching $y = x^2 - 6x$ and $y = 2x - 16$.

2 Consider the line $y = -2x + k$ and the parabola $y = x^2 - 6x$.

a Show that the x-coordinates of the points of intersection satisfy the equation $x^2 - 4x - k = 0$.

$-2x + k = x^2 - 6x$

$\therefore x^2 - 4x - k = 0$

b Suppose that the line is a tangent to the parabola; in other words, the two roots of the quadratic equation are equal. Let the roots be α and α.

i Using sum of roots, show that $\alpha = 2$.

$\alpha + \alpha = -\frac{b}{a}$

$2\alpha = 4$

$\alpha = 2$

ii Using product of roots, show that $\alpha^2 = -k$ and hence find k.

$\alpha \times \alpha = \frac{c}{a}$

$\alpha^2 = -k$

$2^2 = -k$

$k = -4$

iii Therefore, state the equation of the tangent line and find its point of contact with the parabola.

Tangent line: $y = -2x - 4$

Point of intersection: $x^2 - 4x + 4 = 0$

$(x - 2)^2 = 0$

$x = 2$

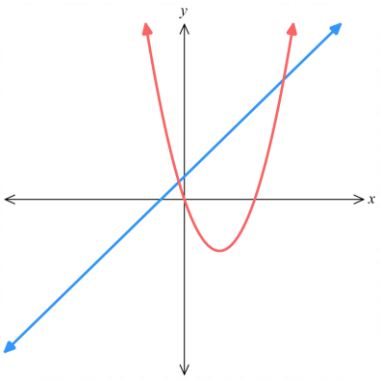
$y = -2(2) - 4 = -8$

Tangent point at $(2, -8)$

c Use DESMOS to visualise what is happening by sketching $y = -2x - 4$ and $y = x^2 - 6x$.

3 The line $y = x + 1$ meets the parabola at $y = x^2 - 3x$ at points A and B.

A sketch has been provided.



a Show that the x-coordinates α and β of A and B satisfy $x^2 - 4x - 1 = 0$.

$x + 1 = x^2 - 3x$

$x^2 - 4x - 1 = 0$

b Find $\alpha + \beta$, and hence find the coordinates of the midpoint M of AB.

$x = \frac{4 \pm \sqrt{(-4)^2 - 4(1)(-1)}}{2}$

$x = \frac{4 \pm \sqrt{20}}{2}$

$= \frac{4 \pm 2\sqrt{5}}{2}$

$= 2 \pm \sqrt{5}$

$\alpha = 2 + \sqrt{5}, \beta = 2 - \sqrt{5}$

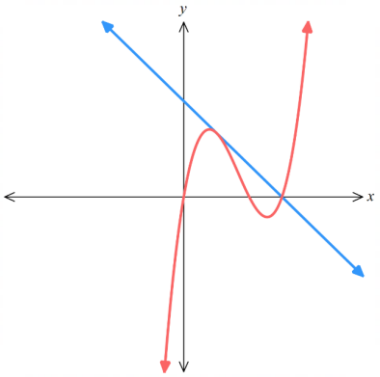
Midpoint of x coordinates is $\frac{2+\sqrt{5}+2-\sqrt{5}}{2} = 2$

Substitute into equation of the line (NOT the parabola): $y = 2 + 1 = 3$

$\therefore (2,3)$

4 Consider the line $y = 3 - x$ and the cubic $y = x^3 - 5x^2 + 6x$.

A sketch has been provided.



a Show that the x-coordinates of the points of intersection satisfy $x^3 - 5x^2 + 7x - 3 = 0$.

$3 - x = x^3 - 5x^2 + 6x$

$\therefore x^3 - 5x^2 + 7x - 3 = 0$

b Show that $x = 1$ and $x = 3$ are zeroes of the equation, and hence use the sum of zeroes to find the third root.

$P(1) = 0$

$P(3) = 0$

Let third root be α

$1 + 3 + \alpha = 5$

$\alpha = 1$

c Explain why the line is tangent to the cubic, then find the points of intersection.

1 is a double root, so the line is tangent to the parabola at $x = 1$

Points of intersections are $(1,2)$ and $(3,0)$

5

a Show that the x-coordinates of the points of intersection of the line $y = mx$ and the cubic $y = x^3 - 5x^2 + 6x$ satisfy the equation $x^3 - 5x^2 + (6 - m)x = 0$

$mx = x^3 - 5x^2 + 6x$

$x^3 - 5x^2 + 6x - mx = 0$

$x^3 - 5x^2 + (6 - m)x = 0$

b Explain why one of the points of intersection is at $x = 0$.

$P(0) = 0$

Alternatively, you can factor out $x(x^2 - 5x + 6 - m) = 0$

$\therefore$ by null factor law, $x = 0$ is a zero.

c Suppose that the line is tangent to the cubic at a point other than the origin, so that the roots of the equation are $0, \alpha, \alpha$.

i Using sum of roots, show that $\alpha = \frac{5}{2}$

$2\alpha = 5$

$\alpha = \frac{5}{2}$

ii Using the product of pairs of roots, show that $\alpha^2 = 6 - m$, and hence find m.

$0 + 0 + \alpha^2 = 6 - m$

$\alpha^2 = 6 - m$

$m = 6 - \alpha^2$

$= 6 - \left(\frac{5}{2}\right)^2$

$= -\frac{1}{4}$

iii Find the equation of the tangent and its point of contact T.

Equation of tangent: $y = -\frac{x}{4}$

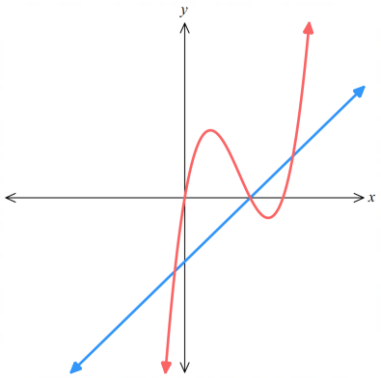
Point of contact at $x = \frac{5}{2}$, $y = -\frac{\left(\frac{5}{2}\right)}{4} = -\frac{5}{8}$

$\left(\frac{5}{2}, -\frac{5}{8}\right)$

d Use DESMOS to visualise what is happening by sketching the line and the cubic.

6 The line $y = x - 2$ meets the cubic $x^3 - 5x^2 + 6x$ at $F(2,0)$ and also at A and B.

a Show that the x-coordinates α and β of A and B satisfy $x^3 - 5x^2 + 5x + 2 = 0$



$x^3 - 5x^2 + 6x = x - 2$

$\therefore x^3 - 5x^2 + 5x + 2 = 0$

b Find $\alpha + \beta$.

Using sum of roots,

$\alpha + \beta + 2 = 5$

$\alpha + \beta = 3$

c Hence the coordinates of the midpoint M of AB.

Averaging the x-coordinates of A and B will find the x-coordinate of the midpoint; substitute this into the equation of the line to find:

$\left(\frac{3}{2}, -\frac{1}{2}\right)$

d Show that $\alpha\beta = -1$

Using product of roots,

$2\alpha\beta = -2$

$\alpha\beta = -1$

e Use the identity $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$ to show that the horizontal distance $|\alpha - \beta|$ between A and B is $\sqrt{13}$.

$(\alpha - \beta)^2 = (3)^2 - 4(-1) = 13$

$\therefore |\alpha - \beta| = \sqrt{13}$

f Hence find the length of AB.

The line has gradient 1, so if the horizontal distance is $\sqrt{13}$, the vertical distance is also $\sqrt{13}$.

Alternatively, you could find y-coordinates of the points of intersection by substituting x-values into the equation of the line.

$AB = \sqrt{\left(\sqrt{13}\right)^2 + \left(\sqrt{13}\right)^2}$

$AB = \sqrt{26}$